

Planes and Distance in 3D

Trim Ch 11, S. 11.5 & 11.6

Outline:

1. Planes as Vector Functions
2. Distance in 3D



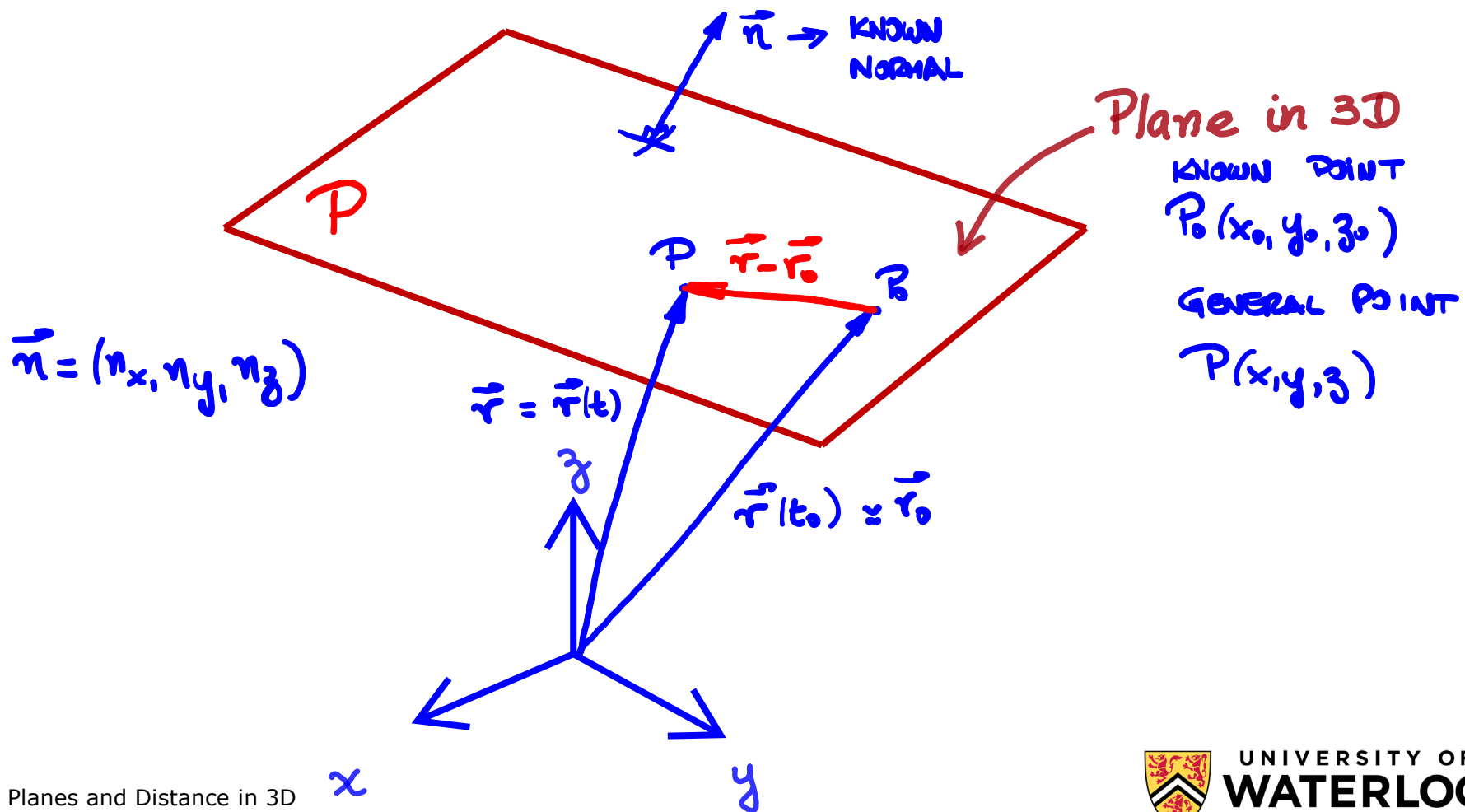
<https://www.dronezon.com/>



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Planes in 3D – Trim S.11.5

How can we find a general equation for a plane in 3D using vector functions?



Vector form of the equation of a plane

Noticing that $(\vec{r} - \vec{r}_0)$ lies completely in the plane and that $\vec{n}$ is perpendicular to the plane, we can use vector analysis to write,

$$(\vec{r} - \vec{r}_0) \cdot \vec{n} = 0 \quad \dots (1)$$

EQUATING (1)

$$(\vec{r} \cdot \vec{n}) - \underbrace{(\vec{r}_0 \cdot \vec{n})}_{\text{CONSTANT}} = 0$$

$$\vec{r} \cdot \vec{n} = \vec{r}_0 \cdot \vec{n} = x_0 n_x + y_0 n_y + z_0 n_z = \phi$$

$$\boxed{\vec{r} \cdot \vec{n} = \phi}$$

VECTOR EQ'
OF A PLANE

Other forms of the equation of a plane

Recalling the vector equation we just derived for a plane in 3D:

$$(\vec{r} - \vec{r}_0) \cdot \vec{n} = 0$$

Other forms of the equation of the plane can be found if we expand this equation as:

$$((x - x_0), (y - y_0), (z - z_0)) \cdot (n_x, n_y, n_z) = 0$$

General form of the equation of a plane:

$$n_x(x - x_0) + n_y(y - y_0) + n_z(z - z_0) = 0$$

GENERAL
FORM EQ'
OF PLANE

Other forms of the equation of a plane

Scalar form of the equation of a plane:

Re-arranging the terms on the general form of the equation of the plane,

$$n_x x + n_y y + n_z z = \underbrace{n_x x_0 + n_y y_0 + n_z z_0}_{\phi}$$

$n_x x + n_y y + n_z z = \phi$

A
↗

B
↗

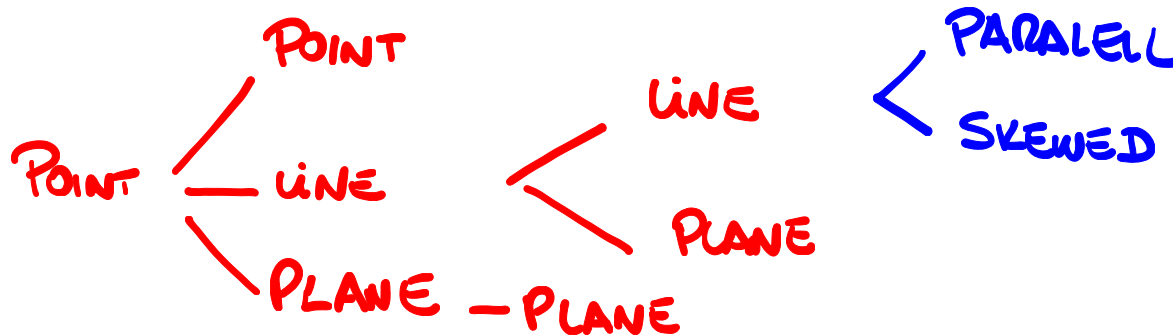
C
↗

D
↗

$$\vec{n} = (n_x, n_y, n_z)$$

Distance in 3D – Trim S.11.6

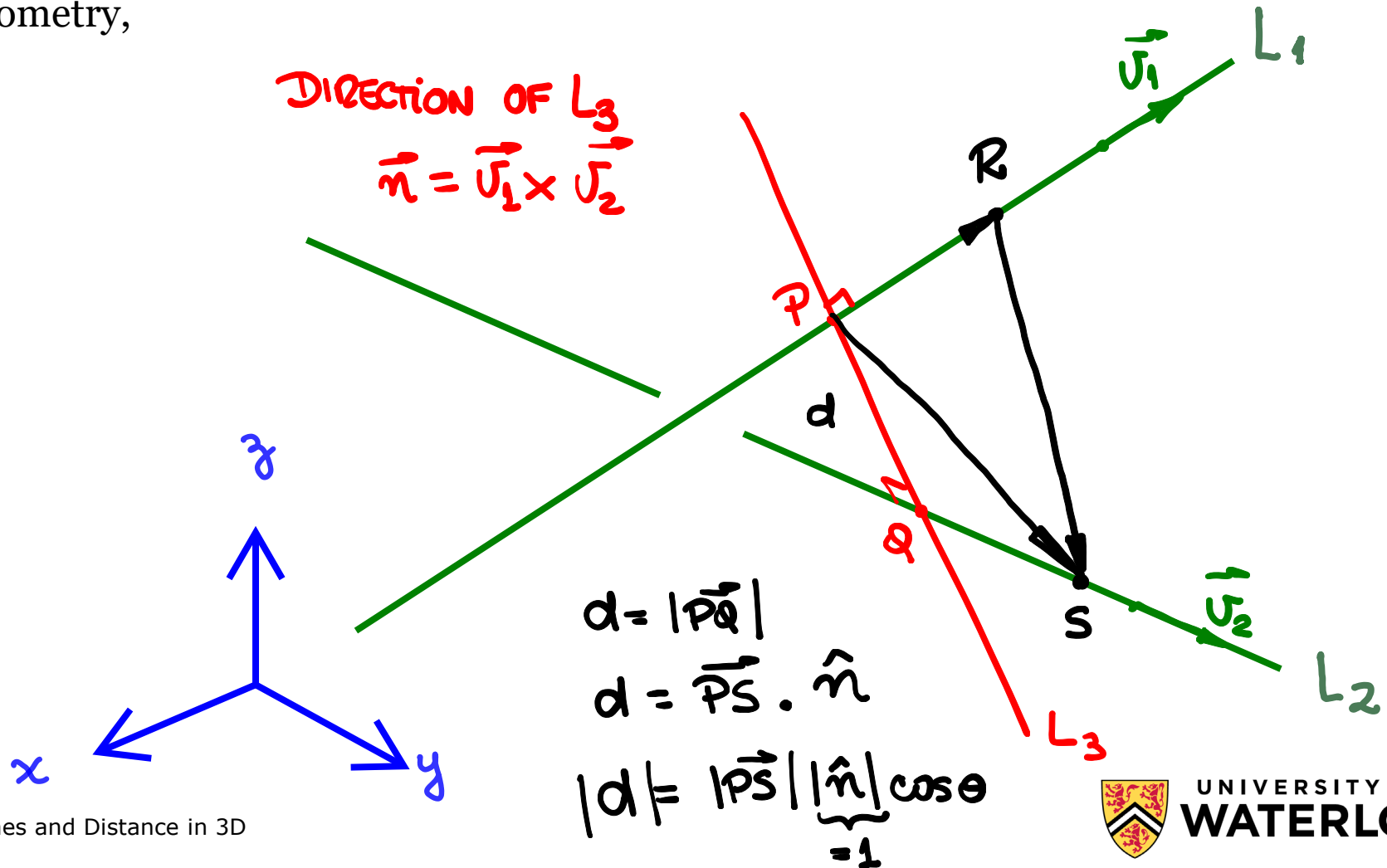
There are six distances of interest between geometric objects in space:



Each type of distance has unique properties. The calculation of these distances are important for multiple applications such as path planning, collision detection, and modeling spatial relationships, among others.

Distance between skewed lines

Skewed lines are lines that do not intersect and are not parallel. Let us look at their geometry,



Distance between skewed lines

Now that we have determined the geometry of the problem, let us use vector analysis to calculate the distance.

$$d = |\vec{PQ}| = |\vec{PS} \cdot \hat{n}|$$

$$\vec{PS} = \vec{PR} + \vec{RS}$$

$$d = |(\vec{PR} + \vec{RS}) \cdot \hat{n}|$$

$$d = |\cancel{\vec{PR} \cdot \hat{n}} + \vec{RS} \cdot \hat{n}|$$

$$d = \vec{RS} \cdot \hat{n}$$

VECTORIAL
FORMULATION !

Distance between skewed lines

Vectorial formulation for the distance between skew lines:

$$d = |\overrightarrow{RS} \cdot \hat{n}|$$

For numerical calculations:

1. The points R and S are arbitrary and can be obtained from each one of the line equations L_1 and L_2 , respectively.
2. The direction of the normal of the two lines is given by,

$$\vec{n} = \vec{v}_1 \times \vec{v}_2$$

$$\hat{n} = \frac{\vec{n}}{|\vec{n}|}$$

A numerical problem that uses the distance between two skewed lines will be solved in Tutorial 2



Distance Between a Point and a Line

Problem:

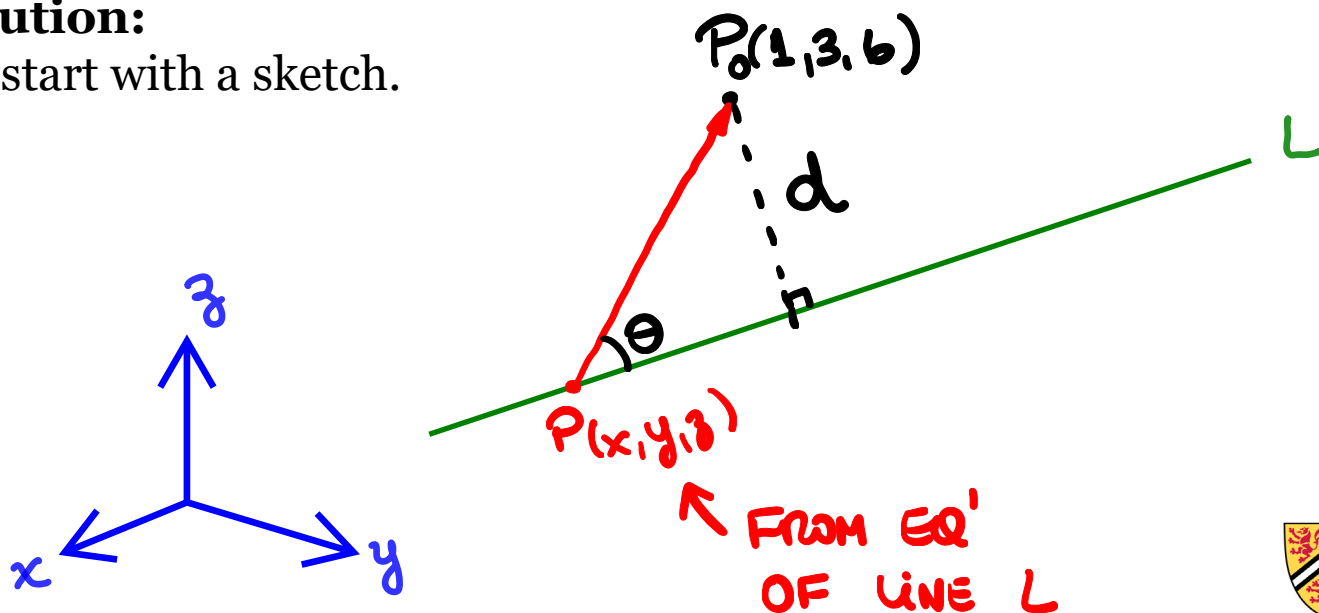
Suppose a drone is flying across the line,

$$x - 1 = \frac{y - 2}{2} = \frac{z - 4}{3}$$

How close does the drone pass the point $P_0(1,3,6)$?

Solution:

We start with a sketch.

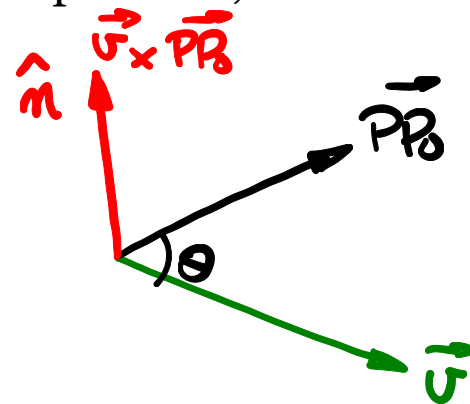


Distance Between a Point and a Line

Now that we have determined the geometry of the problem, let us use vector analysis to calculate the distance.

$$d = \underbrace{|\vec{P}\vec{P}_0| \sin \theta}$$

$$\vec{U} \times \vec{P}\vec{P}_0 = |\vec{U}| |\vec{P}\vec{P}_0| \sin \theta \cdot \hat{n}$$



TAKING MAGN. ON BOTH SIDES,

$$|\vec{U} \times \vec{P}\vec{P}_0| = |\vec{U}| \underbrace{|\vec{P}\vec{P}_0| \sin \theta}_d |\underbrace{\hat{n}}_{=1}|$$

$$d = \frac{|\vec{U} \times \vec{P}\vec{P}_0|}{|\vec{U}|} \quad \dots (\text{I})$$



Distance Between a Point and a Line

IN OUR PROBLEM, $\rightarrow$ TRY THIS YOURSELF FIRST!

$$\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-4}{3}$$

CONVERTING TO PARAMETRIC FORM

$$\begin{cases} x = 1 + t \\ y = 2 + 2t \\ z = 4 + 3t \end{cases}$$

$$\vec{v} = (1, 2, 3)$$

$$\text{FOR } t=0 \rightarrow P(1, 2, 4)$$

$$\text{THEN, } \vec{PR_0} = (1, 3, 6) - (1, 2, 4) = (0, 1, 2)$$

IN (I),

$$d = \frac{|(0, 1, 2) \times (1, 2, 3)|}{|(1, 2, 3)|} = \dots = \frac{\sqrt{21}}{7}$$



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General methodology to calculate distance:

- Sketch
- Determine the vector geometry of the problem
- Use vector analysis to determine the vector formulation
- Perform your numerical calculations

Summary

- We have determined the equation of a plane in 3D in vector, general and scalar form
- There are six distances of interest between geometric objects in space. Distances can be calculated for any of these six combinations.
- We have determined a general procedure that we can use for the calculation of any distance in 3D



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Thank You